

Supplementary Material for NEURA: Local Excitable Routing Control for Stress-Adaptive Networks

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1. Additional Service and Safety Results

This supplementary material provides the supporting evidence chain for the low-amplification operating point studied in the main paper. It reports additional startup, path-stretch, topology-family, gray-failure, safety, mechanism-attribution, and parameter-sensitivity results. Together, these results connect the main delivery-control tradeoff to startup usability, bounded detours, topology robustness, routing-safety indicators, mechanism roles, and parameter robustness.

1.1. Startup Convergence and Path Stretch

The startup and stretch experiment examines two related questions: how much startup convergence is required before the network becomes fully reachable, and whether the low-overhead operating point is obtained by inflating data-plane routes. Fig. 1 reports startup convergence together with burst-window path stretch at the representative 3× hotspot workload.

NEURA reaches full reachability in 40 ms on average, faster than 88 ms for OSPF-TE and 62 ms for Bandit, and matching Triggered-TE at 40 ms. Its startup control budget is also lower, averaging about 1.76 MB before full reachability, compared with 2.73 MB for OSPF-TE, 3.36 MB for Triggered-TE, and 3.22 MB for Bandit. Under the representative 3× hotspot workload, NEURA reaches a mean burst-window path stretch of about 1.048, compared with 1.007 for OSPF-TE, 1.031 for Triggered-TE, and 1.002 for Bandit. These results show that the low-amplification operating point retains fast startup usability and avoids excessive route inflation.

1.2. Topology-Family Checks

Topology-family checks were run on connected 100-node Barabási–Albert (BA-100) and random geometric (RGG-100) graphs to test whether the locality pattern depends on ER-style path mixing. In the BA-100 locality check, NEURA emits about 83.4 hotspot-window updates and keeps all of them within two hops of the stressed region, with a 90th percentile emission radius of one hop. The matched OSPF-TE periodic check emits about 500 updates, has a 90th percentile radius of 2.6 hops, and spreads a larger share of activity beyond the immediate neighborhood.

The stricter RGG-100 locality check gives the same qualitative result. NEURA emits about 36.2 hotspot-window updates,

keeps all activity within two hops, and has no measured activity beyond five hops. OSPF-TE emits about 420 updates, keeps only about 27.6% of hotspot-window updates within two hops, and places about 38.8% of activity at five or more hops. On the RGG-100 3× stress sweep, NEURA preserves 51.9% burst-window delivery with 14.86 MB of control traffic and about 181 route changes per node, while OSPF-TE preserves 55.5% with 72.97 MB and about 664 route changes per node. Triggered-TE preserves 50.5% with 75.66 MB and about 210 route changes per node. Thus, changing the topology family changes the absolute service level, especially in spatially constrained graphs, but preserves the low-amplification ordering.

1.3. Gray-Failure Validation

The gray-failure validation keeps the topology unchanged and instead degrades the egress service rate of one transit node to 30% of nominal capacity for 300 ms. This scenario tests whether queue-derived local stress is sufficient to trigger route adaptation without a hard link-down event.

Table 1: Gray-failure validation. The impaired transit node remains connected while its egress service rate is reduced to 30% of nominal capacity for 300 ms.

Method	Delivery (%)	Ctrl/epoch (KB)	Total ctrl. (MB)
NEURA	92.5	20.3	6.39
Triggered-TE	96.4	329.3	41.68
OSPF-TE	97.3	378.3	44.53
Bandit	97.1	630.7	74.43

NEURA reaches a mean impairment-window delivery ratio of 92.5%, lower than the stronger engineering and learning-based baselines, but it does so with a sharply reduced control budget. Triggered-TE closes much of the service gap, but its control budget remains far closer to the periodic baseline than to NEURA. These results support the interpretation that queue-derived local stress is sufficient to trigger useful adaptation under gray failure while preserving the low-control operating point.

1.4. Safety Indicators

Table 2 separates routing-correctness effects from the control/service tradeoff. The reported values are computed after each method first reaches full pairwise reachability. Pairwise reachability measures whether each source destination pair has

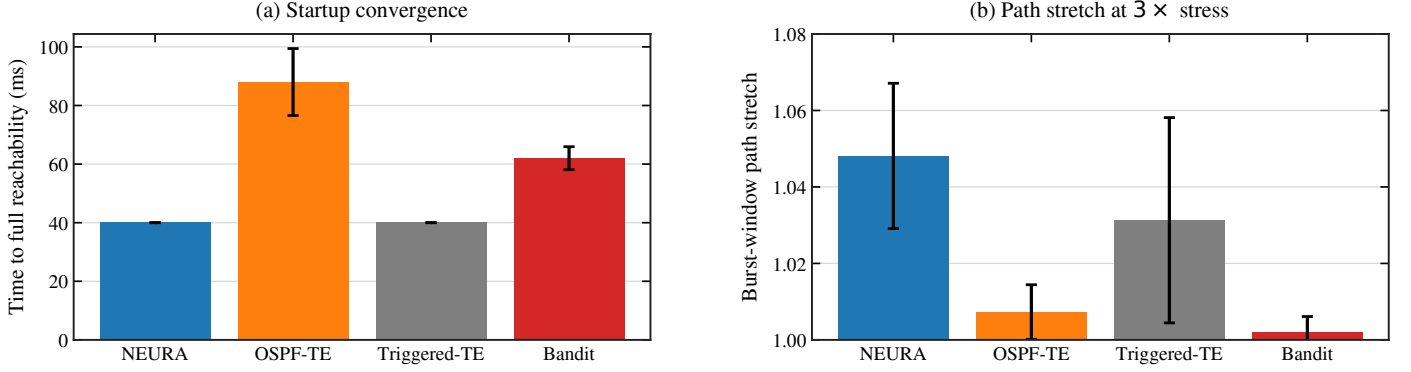


Figure 1: Startup cost and data-plane path penalty. Panel (a) reports the time needed to establish full reachability from startup. Panel (b) reports mean burst-window path stretch at the representative $3\times$ hotspot workload. NEURA matches the fastest startup convergence and maintains bounded path inflation rather than achieving low control traffic by allowing arbitrarily poor routes.

at least one usable forwarding chain at the observation instant. The node-complete-route ratio is stricter: it measures the fraction of nodes whose forwarding table contains a currently valid next hop for every destination after local validity checks. A low complete-route ratio can therefore coexist with high burst-window delivery when the missing or stale entries do not lie on the active traffic demands. For NEURA and the stronger periodic, multipath, and learning baselines, final pairwise reachability returns to 100% in these representative runs and the final node-complete-route ratio also returns to 100%.

The nearest-neighbor Triggered-TE baseline is the main exception. It preserves strong burst-window delivery in the main service figures, but its post-startup reachability floor drops under the representative hotspot and continuous-chaos workloads, and its worst final node-complete-route ratio is much lower than the other baselines. This does not mean that Triggered-TE is unusable in the service-window measurements; it means that, under the stricter all-destinations table-completeness check, some inactive or noncritical destination entries remain stale or invalid after stress. By contrast, the main NEURA tradeoff is not explained by persistent loops, blackholes, or incomplete forwarding-table recovery.

2. Mechanism Attribution

The mechanism attribution study asks which parts of the control law are responsible for the observed operating point. Table 3 combines two targeted studies that use the same routing engine but isolate different failure modes of the control plane.

The top row studies rebound after a two-stage disturbance. In the minimal event-driven core without memory, the rebound ratio after release is about 1.57% and the system continues to accumulate about 21 post-stage-2 route changes. Adding memory alone reduces the rebound ratio to about 1.40% and eliminates post-stage-2 route changes. The full mechanism lowers the rebound ratio further to about 0.28% while also reducing total emitted updates from about 208k to about 166k over the same run.

The bottom row studies switch suppression inside the queue-closed-loop continuous-chaos workload. Removing the switch-

suppression terms increases route rewrites from about 100 to about 126 per node and raises the peak event rate from about 29.7k to about 41.6k messages per tick. Mean burst delivery remains slightly higher without switch suppression, which clarifies the role of this mechanism: it suppresses excessive local switching and prevents a stressed region from turning into a sharper control-plane burst.

3. Parameter Sensitivity

The focused sensitivity study indicates that the reported operating point lies inside a usable tested neighborhood. Threshold θ and refractory duration T_{ref} form the clearest response-versus-overhead tradeoff. Increasing θ from 1.1 to 1.7 changes the mean burst-delivery ratio from 0.925 to 0.913 while reducing mean total emitted updates from 2.36×10^5 to 1.78×10^5 . Increasing T_{ref} from 2 to 6 keeps the mean burst-delivery ratio near 0.923 to 0.926 while reducing mean total emitted updates from 2.24×10^5 to 1.62×10^5 .

The memory-retention and slow-timescale sweeps are comparatively stable in the tested region. Varying η from 0.55 to 0.90 changes the mean burst-delivery ratio only from 0.924 to 0.922, and varying ω_s from 0.15 to 0.55 changes it only from 0.923 to 0.923. In the repeated-disturbance sensitivity sweep, post-stage-2 route changes remain saturated at zero across the tested values.

Data and Code Availability

The code and curated artifacts used for the supplementary analyses are available in the public NEURA reproduction repository at <https://github.com/unicorners-compin/neura-routing-artifact>. The internal package name `snn_sra_sim` is a development identifier for the NEURA implementation.

Table 2: Post-startup safety indicators on ER-100 across five seeds. Values report the worst case over seeds after full reachability is first established. Complete-route ratio is a stricter forwarding-table completeness metric than pairwise reachability.

Scenario	Method	Worst post-startup reach. (%)	Worst loop (%)	Worst blackhole (%)	Worst final reach. (%)	Worst final complete-route (%)
Hotspot 3×	NEURA	99.87	0.13	0.00	100.0	100.0
Hotspot 3×	Triggered-TE	98.97	1.03	0.00	99.05	6.0
Hotspot 3×	OSPF-TE	99.88	0.12	0.08	100.0	100.0
Hotspot 3×	TE+ECMP	99.92	0.00	0.08	100.0	100.0
Hotspot 3×	Bandit	99.96	0.00	0.04	100.0	100.0
Continuous chaos	NEURA	99.87	0.13	0.00	100.0	100.0
Continuous chaos	Triggered-TE	97.64	2.36	0.00	99.02	3.0
Continuous chaos	OSPF-TE	99.88	0.12	0.08	100.0	100.0
Continuous chaos	TE+ECMP	99.92	0.00	0.08	100.0	100.0
Continuous chaos	Bandit	99.96	0.00	0.04	100.0	100.0
Gray failure	NEURA	99.94	0.06	0.00	100.0	100.0
Gray failure	Triggered-TE	99.11	0.89	0.00	99.12	13.0
Gray failure	OSPF-TE	99.92	0.00	0.08	100.0	100.0
Gray failure	TE+ECMP	99.92	0.00	0.08	100.0	100.0
Gray failure	Bandit	99.96	0.00	0.04	100.0	100.0

Table 3: Mechanism attribution for NEURA. Panel A isolates rebound suppression, and Panel B isolates burst containment. Bold entries mark the decisive attribution values discussed in the text.

Variant	Del. (%)	Rebound (%)	Post-stage-2 changes	Updates (k)
A. Rebound suppression				
Minimal core	96.0	1.57	21.0	208.4
Memory only	96.0	1.40	0.0	180.8
Full NEURA	96.0	0.28	0.0	165.5
Variant	Burst del. (%)	Route chg. per node	Peak rate (10^3 /tick)	Loss ticks
B. Burst containment				
Full NEURA	96.0	100.1	29.7	7.4
No memory	96.0	100.0	29.7	7.4
No switch supp.	96.7	125.9	41.6	8.0
No slow state	96.0	100.1	29.7	7.4

Table 4: Sensitivity-summary of the main control parameters in NEURA. The table reports the dominant trend observed over each sweep rather than the full sampled matrix.

Parameter	Range	Main effect	Interpretation
θ	1.1–1.7	Delivery 0.925 $\rightarrow$ 0.913; updates $2.36 \times 10^5 \rightarrow 1.78 \times 10^5$	Higher threshold lowers control overhead at a modest service cost
T_{ref}	2–6	Delivery nearly unchanged; updates $2.24 \times 10^5 \rightarrow 1.62 \times 10^5$	Temporal suppression effectively reduces control activity
η	0.55–0.90	Delivery changes only from 0.924 to 0.922	Memory retention is stable in the tested region
ω_s	0.15–0.55	Delivery remains essentially unchanged at 0.923	Slow state acts as a secondary stabilizer rather than a main lever
g_0	0.25–0.65	Delivery 0.925 $\rightarrow$ 0.923; chaos control 7.94 $\rightarrow$ 8.11 MB	Stable operating region under moderate guard variation
g_1	0.15–0.55	Route changes 110 $\rightarrow$ 101; delivery 0.928 $\rightarrow$ 0.918	A smooth churn–service knob
w_n	0.2–0.5	Reducing w_n to 0.2 lowers delivery to 0.887	Explicit next-hop stress awareness is important
TTL	4–12	No measurable effect	No measurable effect in the tested locality range